glance back at the following section, which summarizes the key insights of the second superstring revolution.

A Summary of the Second Superstring Revolution

The primary insight of the second superstring revolution is summarized by Figures 12.1 and 12.2. In Figure 12.1 we see the situation prior to the recent ability to go (partially) beyond the approximation methods physicists have traditionally used to analyze string theory. We see that the five string theories were thought of as being completely separate. But, with the newfound insights emerging from recent research, as indicated in Figure 12.2, we see that, like the starfish's five arms, all of the string theories are now viewed as a single, all-encompassing framework. (In fact, by the end of this chapter we will see that even a sixth theory—a sixth arm-will be merged into this union.) This overarching framework has provisionally been called M-theory, for reasons that will become clear as we proceed. Figure 12.2 represents a landmark achievement in the quest for the ultimate theory~ Seemingly disconnected threads of research in string theory have now been woven together into a single tapestry—a unique, all-encompassing theory that may well be the long-sought theory of everything.

Although much work remains to be done, there are two essential features of M-theory that physicists have already uncovered. First, M-theory has eleven dimensions (ten space and one time). Somewhat as Kaluza found that one additional spatial dimension allowed for an unexpected merger of general relativity and electromagnetism, string theorists have realized that one additional spatial dimension in string theory–beyond the nine space and one time dimensions discussed in preceding chapters—allows for a deeply satisfying synthesis of all five versions of the theory. Moreover, this extra spatial dimension is not pulled out of thin air; rather, string theorists have realized that the reasoning of the 1970s and 1980s that led to one time and nine space dimensions was approximate, and that exact calculations, which can now be completed, show that one spatial dimension had hitherto been overlooked.

The second feature of M-theory that has been discovered is that it contains vibrating strings, but it also includes other objects: vibrating two-dimensional membranes, undulating three-dimensional blobs